

NanoGT Match Report: Usher syndrome type 1B

Disease: Usher syndrome type 1B (ORPHA:886)

Primary gene: MYO7A

Gene CDS: 6648 bp

Inheritance: Autosomal recessive

Target tissues scored: retina, cochlea

Interpretation

- At least one high-confidence precedent was found, but this is still a precedent match rather than a clinical-trial recommendation.
- Main review flags: Vector does not naturally cover all annotated disease tissues: cochlea; Cell-autonomous protein across multiple tissues; high transduction coverage may be required; Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility.

Disease Mechanism Evidence

Molecular mechanism: loss of function

Mechanistic detail: MYO7A myosin VIIA deficiency causing combined deafblindness; large CDS (~6.6 kb) exceeds single-AAV capacity

Gene-addition compatibility: conditional

Preferred modality class: dual aav or lv retinal cochlear delivery

Evidence level/status: direct / needs_user_fact_check

Evidence summary: MYO7A LOF supports gene addition logic but oversized CDS requires dual-AAV or lentiviral strategies for retinal and cochlear delivery

Evidence source: OMIM MYO7A 276900

Study-Level Limitations

- Catalog-relative ranking: current catalog contains 21 precedent programs and 8 vectors, so absence of a strong match is not proof that no therapy is possible.
 - Modality coverage is limited mainly to AAV and lentiviral precedents; dual-AAV, LNP/mRNA, genome editing, ASO, and transplant-enabling strategies are not fully represented.
 - Few catalog vectors cover the annotated disease tissue(s): AAV2.
 - Endpoint readiness: retinal diseases often have measurable endpoints such as OCT, ERG, visual acuity, visual fields, or mobility testing, but genotype-specific progression still needs confirmation.
 - Endpoint risk: multi-system disease may need a hierarchy of primary and secondary endpoints; one tissue response may not equal whole-disease benefit.
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Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	Luxturna	AAV2	7.5/10	High	approved
2	Strimvelis	LV	7.0/10	Medium	approved
3	CPCB-RPE1	AAV8	6.9/10	Medium	phase2/3

Rank	Program	Vector	Score	Confidence	Approval
4	BMN 307	AAV5	6.8/10	Medium	phase2
5	GS010	AAV2	6.6/10	Medium	phase3

Match #1: Luxturna

Precedent disease: Leber congenital amaurosis type 2

Vector: AAV2

Tissue target: retina/RPE

Composite score: 7.5 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	0.50	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.80	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	7.52	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 2646bp / cargo 4700bp (56% utilized)
- Vector tropism plus precedent target match: retina
- Both intracellular proteins

- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: retinal_visual_cycle
- Disease mechanism: loss of function — MYO7A myosin VIIA deficiency causing combined deaf-blindness; large CDS (~6.6 kb) exceeds single-AAV capacity
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: MYO7A LOF supports gene addition logic but oversized CDS requires dual-AAV or lentiviral strategies for retinal and cochlear delivery
- Mechanism source: OMIM MYO7A 276900 (<https://omim.org/entry/276900>)
- Approval status: approved
- Vector immunogenicity (AAV2): very high (~55%) — majority of patients may be ineligible; major trial design challenge
- Narrow therapeutic window — neonatal onset requires delivery within weeks of birth; trial design must incorporate newborn screening programmes
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF-2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010
- ORGANELLE TARGETING: COMPATIBLE — MYO7A standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Vector does not naturally cover all annotated disease tissues: cochlea
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Gene annotation incomplete; packaging, protein class, and localization scores need manual confirmation
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #2: Strimvelis

Precedent disease: ADA-SCID

Vector: LV

Tissue target: hematopoietic

Composite score: 7.0 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.80	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.95	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 1092bp / cargo 8000bp (14% utilized)
- No tissue overlap (disease: ['retina', 'cochlea'], vector: ['hematopoietic'])
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Different pathway (retinal_phototransduction vs immune_hematopoietic)
- Disease mechanism: loss of function — MYO7A myosin VIIA deficiency causing combined deaf-blindness; large CDS (~6.6 kb) exceeds single-AAV capacity
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: MYO7A LOF supports gene addition logic but oversized CDS requires dual-AAV or lentiviral strategies for retinal and cochlear delivery
- Mechanism source: OMIM MYO7A 276900 (<https://omim.org/entry/276900>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Narrow therapeutic window — neonatal onset requires delivery within weeks of birth; trial design must incorporate newborn screening programmes
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more

targeted dose

- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF- 2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010
- ORGANELLE TARGETING: COMPATIBLE — MYO7A standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Vector does not naturally cover all annotated disease tissues: cochlea, retina
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Gene annotation incomplete; packaging, protein class, and localization scores need manual confirmation

Match #3: CPCB-RPE1

Precedent disease: Achromatopsia

Vector: AAV8

Tissue target: retina/photoreceptor

Composite score: 6.9 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.70	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.80	2.0	Can GT be given before irreversible damage?

Dimension	Score	Max	What it measures
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.90	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 2499bp / cargo 4700bp (53% utilized)
- Precedent target match: retina
- Protein class mismatch
- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: retinal_phototransduction
- Disease mechanism: loss of function — MYO7A myosin VIIA deficiency causing combined deaf-blindness; large CDS (~6.6 kb) exceeds single-AAV capacity
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: MYO7A LOF supports gene addition logic but oversized CDS requires dual-AAV or lentiviral strategies for retinal and cochlear delivery
- Mechanism source: OMIM MYO7A 276900 (<https://omim.org/entry/276900>)
- Approval status: phase2/3
- Vector immunogenicity (AAV8): high (~30%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Narrow therapeutic window — neonatal onset requires delivery within weeks of birth; trial design must incorporate newborn screening programmes
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF- 2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010
- ORGANELLE TARGETING: COMPATIBLE — MYO7A standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Vector does not naturally cover all annotated disease tissues: cochlea, retina
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required

- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Gene annotation incomplete; packaging, protein class, and localization scores need manual confirmation
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #4: BMN 307

Precedent disease: Phenylketonuria

Vector: AAV5

Tissue target: liver

Composite score: 6.8 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.60	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.80	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.76	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 1353bp / cargo 4700bp (29% utilized)

- No tissue overlap (disease: ['retina', 'cochlea'], vector: ['liver', 'lung', 'CNS'])
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Different pathway (retinal_phototransduction vs amino_acid_metabolism)
- Disease mechanism: loss of function — MYO7A myosin VIIA deficiency causing combined deaf-blindness; large CDS (~6.6 kb) exceeds single-AAV capacity
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: MYO7A LOF supports gene addition logic but oversized CDS requires dual-AAV or lentiviral strategies for retinal and cochlear delivery
- Mechanism source: OMIM MYO7A 276900 (<https://omim.org/entry/276900>)
- Approval status: phase2
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Narrow therapeutic window — neonatal onset requires delivery within weeks of birth; trial design must incorporate newborn screening programmes
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF- 2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010
- ORGANELLE TARGETING: COMPATIBLE — MYO7A standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Vector does not naturally cover all annotated disease tissues: cochlea, retina
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Gene annotation incomplete; packaging, protein class, and localization scores need manual confirmation
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #5: GS010

Precedent disease: Leber hereditary optic neuropathy

Vector: AAV2

Tissue target: retina/RGC

Composite score: 6.6 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.30	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.80	1.0	Regulatory approval / trial stage
Immunogenicity	0.50	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.80	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.62	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 1378bp / cargo 4700bp (29% utilized)
- Vector tropism plus precedent target match: retina
- Both intracellular proteins
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Different pathway (retinal_phototransduction vs mitochondrial_complex)
- Disease mechanism: loss of function — MYO7A myosin VIIA deficiency causing combined deaf-blindness; large CDS (~6.6 kb) exceeds single-AAV capacity
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: MYO7A LOF supports gene addition logic but oversized CDS requires dual-AAV or lentiviral strategies for retinal and cochlear delivery
- Mechanism source: OMIM MYO7A 276900 (<https://omim.org/entry/276900>)
- Approval status: phase3
- Vector immunogenicity (AAV2): very high (~55%) — majority of patients may be ineligible; major trial design challenge
- Narrow therapeutic window — neonatal onset requires delivery within weeks of birth; trial design must incorporate newborn screening programmes
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must in-

dividually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose

- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF- β 2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010
- ORGANELLE TARGETING: COMPATIBLE — MYO7A standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Vector does not naturally cover all annotated disease tissues: cochlea
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Gene annotation incomplete; packaging, protein class, and localization scores need manual confirmation
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone